Teaching Statement

My teaching style builds on my research on education and is highly entrepreneurial. I have extensive experience teaching both micro- and macroeconomics at the introductory and intermediate level. I have also developed introductory data science courses.

In my research, I have found that there are key bottlenecks that prevent students from reaching their full potential; I target those bottlenecks early in the course to alleviate them. For example, my intermediate microeconomics course is among the most challenging courses undergraduates take in an economics program. It requires heavy use of math and analytical rigor, which many of my students are not confident in at the outset. Building on my research on how people build skills, I dedicate the first few classes to sharpening students' mathematical skills. My classes are not math classes, but establishing this base knowledge allows students to spend more time understanding the material intuitively rather than being lost in the math. So that stronger students are still challenged by the class, I aim high and encourage students to develop an intuitive understanding of the material.

My teaching style is highly entrepreneurial, and I use this to keep students interested. One approach I use is making extensive use of real-world examples. I open classes by having students volunteer to briefly discuss recent economic news. I use this activity to connect economics models to concepts students are passionate or concerned about. One student brought in a news article about the "Mississippi Miracle" where reading test scores in Mississippi jumped from the worst in the US to near the top. This naturally segued into debates about the optimal design of social programs and how the government should spend limited resources. I bring in empirical evidence from the economics literature and my own work to inform these debates. I regularly send out both peer-reviewed papers and less rigorous pop-social science articles and podcasts as supplemental materials following these discussions. Because discussions are student-initiated topics are wide-ranging. Students have brought in

discussions ranging from corporate tax increases to offshore wind farm expansions. By connecting a broad range of topics to the abstract models, students engage more with the class and develop strong intuitions rather than thinking economics is confined to simple prices and quantities.

I used this teaching philosophy while contributing to the design of an interdisciplinary data science course at Sonoma State University. I collaborated with professors from the history and political science departments to design a data science course for humanities majors. The class focuses on designing data collection methodologies and interpreting data driven research students encounter in their every-day lives. We treat understanding of data driven arguments as a liberal arts skill for the 21st century, so we treat understanding and synthesizing data driven writing as being of first-order importance. I brought an economics perspective to the course design by pushing to include more rigorous understanding of the research designs without burying the (mostly humanities) students in math. Together, this makes the course a "great books" course in data science where students learn from and understand well written, data driven works to help them develop their understanding of the world.

I actively encourage students to come to me with questions or research ideas even after finishing my courses. A few promising students have ended up working as an RA for me. I have also deferred several students to RA positions with colleagues of mine if I did not have a need for them at the time. I have closely mentored one particular student who worked as a research assistant for me after finishing one of my classes. With my guidance, he developed a project idea using the knowledge he gained and will be applying to master's programs to continue that research.

Beyond the traditional classroom, I tutor students throughout the year taking a variety of classes. During tutoring sessions, I employ a more Socratic method of instruction than is possible in a large classroom. For example, rather than showing the student how to take a derivative, I will ask them to walk me through how to do it. I let students lead me down

wrong paths so they can figure out for themselves why their idea is wrong, but it is a common misunderstanding. I push students to develop their intuitions by continually asking students to explain the math they are doing in English, not math. Many students are comfortable with the math of economics problems, but not with the intuitions. Many (if not most) of my students do not become professional economists and so will not use the math I teach them in their personal or professional lives. The intuitions, however, will enable them to be informed consumers of news and current events, so I consider building those intuitions the most important part of my job. In sum, all of my pedagogical strategies are dedicated to building students' capacity for learning and applying those skills outside the classroom.